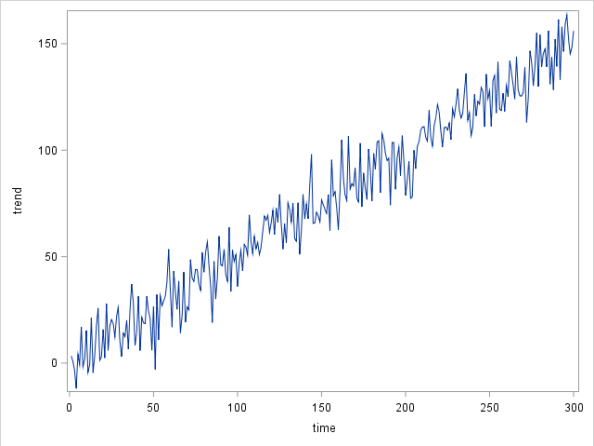
Note 4 Non-Stationary Processes

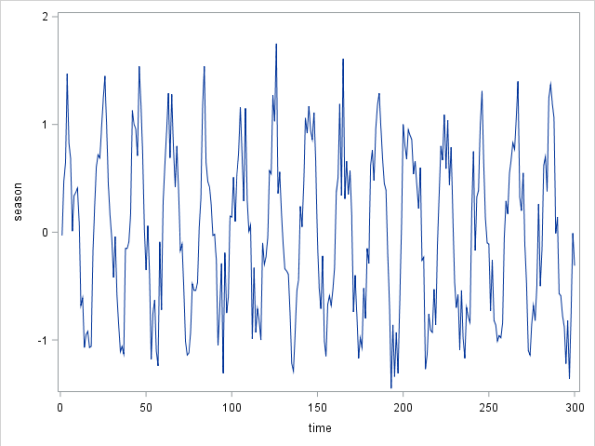
1 Assess Stationarity

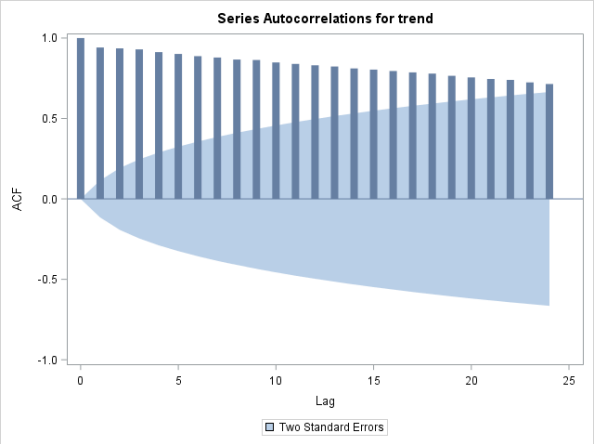
We have discussed in the previous note that we can check if a series is stationary by

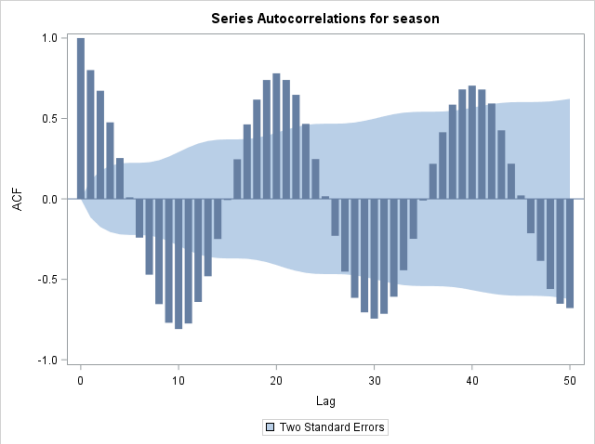
- Plot
- Augmented Dickey-Fuller test

Any time series that is not stationary is a **non-stationary time series**. Some non-stationary patterns can also be recognized from the ACF plot.



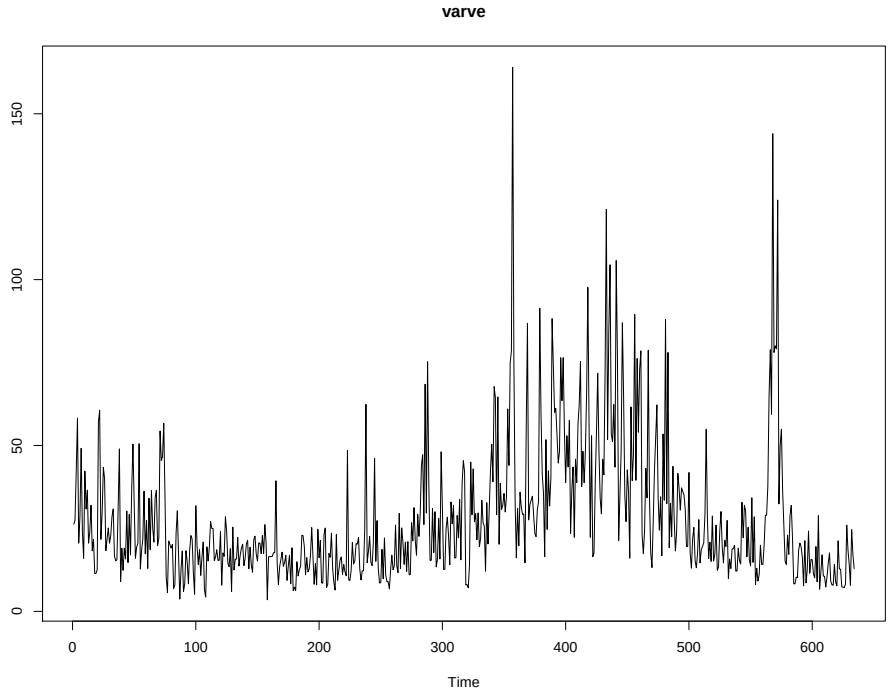






2 Transformation

For a time series to be stationary, the variance shouldn't depend on time t, that is, the magnitude of the series should remain constant over time. This is not always the case in application. Consider the glacier varve thickness from Massachusetts in 634 years below, beginning nearly 12,000 years ago.

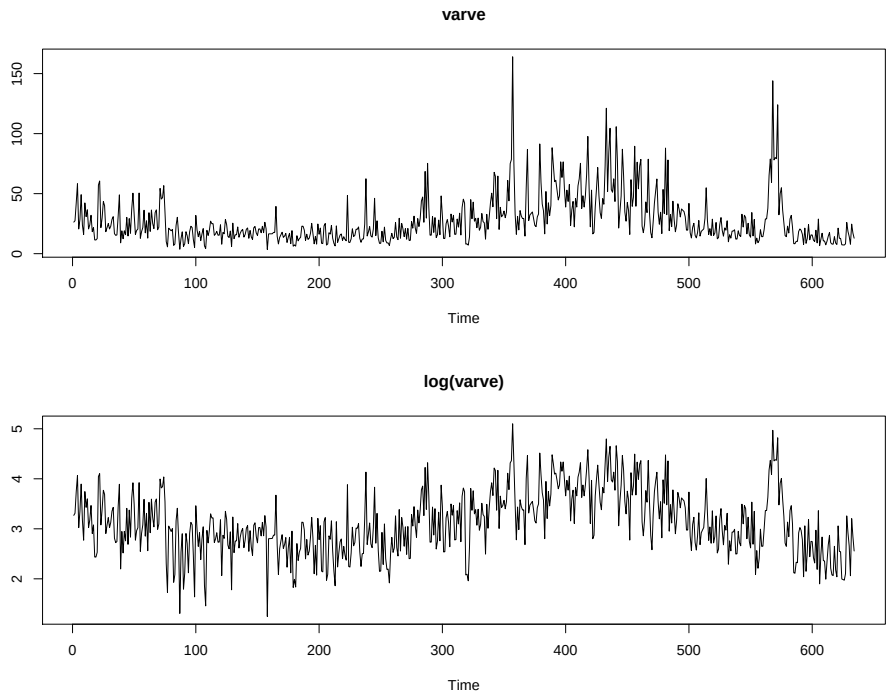


In such cases, transformation may be useful to equalize the variability over the length of a single series. A particular transformation is

$$Y_t = \log(X_t).$$

Other possibilities power transformations such as square root, or the Box-Cox family of the form

$$Y_t = \begin{cases} (X_t^\lambda - 1)/\lambda, & \lambda \neq 0 \\ \log(X_t), & \lambda = 0 \end{cases}$$



3 Classic Decomposition Model

Time series data can exhibit a variety of patterns, and it is often helpful to split a time series into several components, each representing an underlying pattern category. We have discussed two types of time series patterns: **trend** and **seasonality**.

3.1 Additive Decomposition

Let $\{X_t\}$ be a time series, the additive decomposition model is

$$X_t = Tr_t + S_t + \epsilon_t,$$

where Tr_t is the trend component, s_t is the seasonal component with $S_t = S_{t+T}$, and ϵ_t is the residual. The additive decomposition is the most appropriate if the series has approximate constant variance.
Out goal is to estimate the deterministic components Tr_t and S_t such that ϵ_t will turn out to be a stationary time series. Then if there is no correlation among $\{\epsilon_t\}$, we can regard $\{\epsilon_t\}$ as white noise and no further modeling is needed excepting estimating the mean and variance. If there is correlation among $\{\epsilon_t\}$, we need a more complex time series model, and such dependence and assist in predicting future values.

3.2 Multiplicative Decomposition

Alternatively, a multiplicative decomposition model is

$$X_t = Tr_t \times S_t \times \epsilon_t,$$

4 Estimation of the Trend and Seasonal Components

Decomposing a time series is primarily useful for studying time series data, and exploring historical changes over time. The question is how to estimate Tr_t and S_t? Two common approaches are

- Non-parametric approach: linear filter
- Parametric approach: regression

4.1 General Linear Filter

The first step in a classical decomposition is to estimate the trend component via the linear filter.

Let $\{X_t\}$ be a time series, a general linear filter is a linear operation on $\{X_t\}$:

$$\sum_{j=-\infty}^{\infty} a_j X_{t+j},$$

such that $a_j = a_{-j}$ and $\sum_{j=-\infty}^{\infty} a_j = 1$. Here, a_j's are called the weights of the filter.

Some common linear filters are:

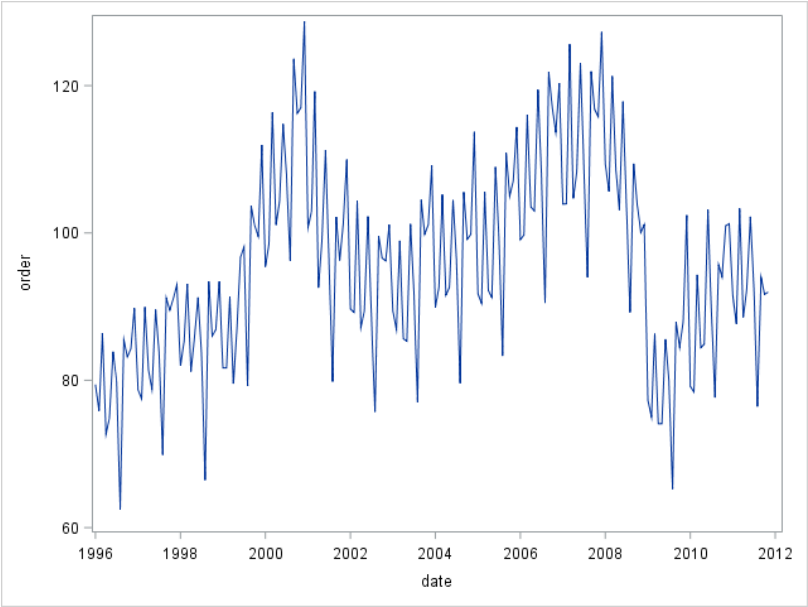
- Moving average filter

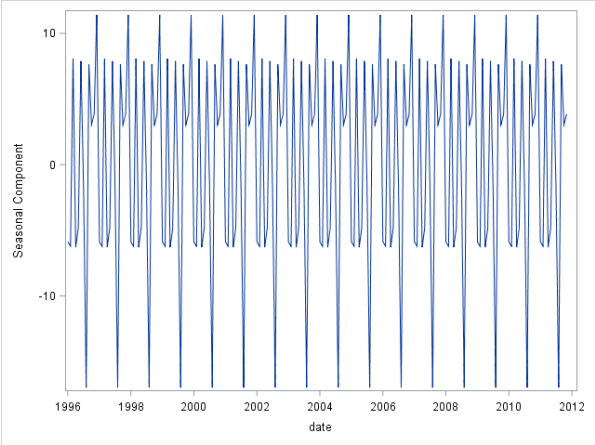
- Binomial filter

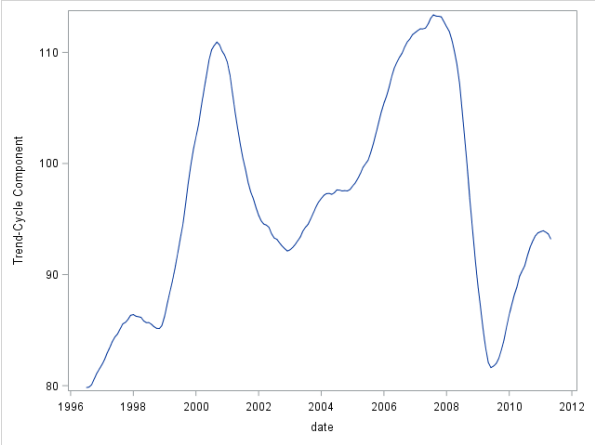
- Exponential filter

Moving average filter is the most commonly used in estimating the trend component.

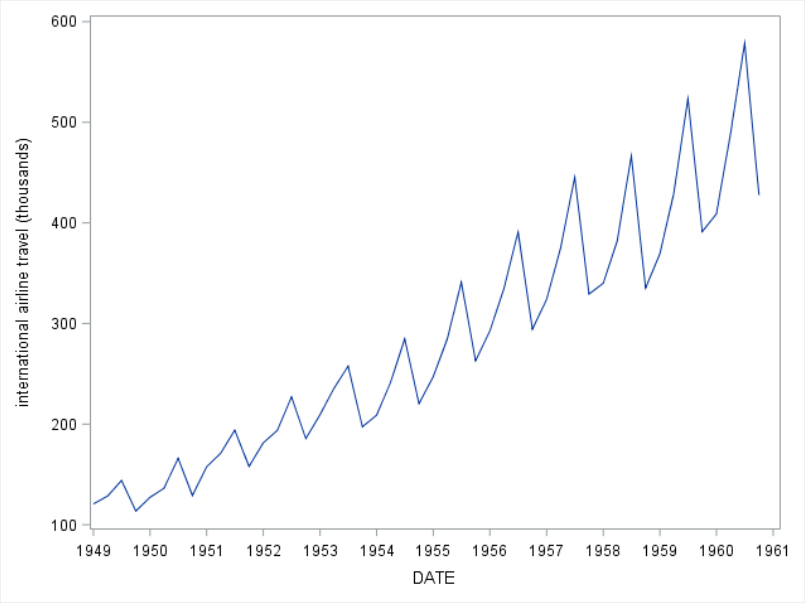
Example 4.1. (Additive model) The data show the number of new orders for electrical equipment (computer, electronic and optical products) in the Euro area (16 countries) from Jan 1996 to Nov 2011.

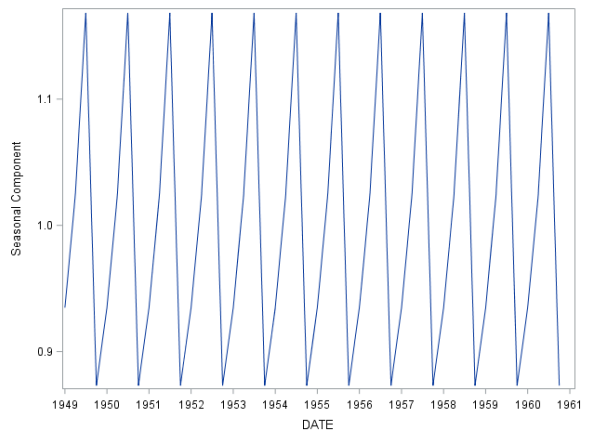


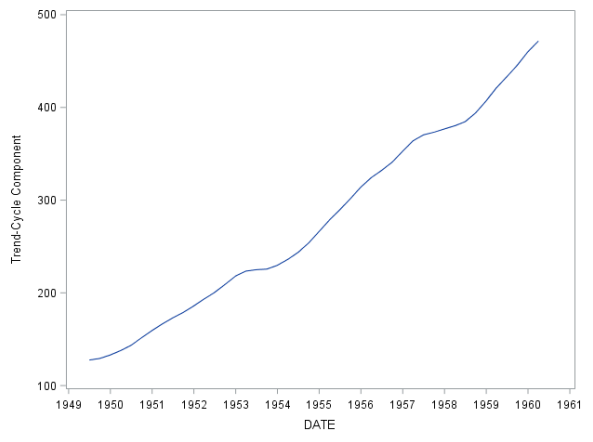




Example 4.2. (Multiplicative model) Below is the quarterly average number of international airline passengers from 1949Q1 to 1960Q4.







4.2 Regression Approach

Another approach to estimate the trend and seasonal component is to fit parametric functions. Common choice for the trend is the polynomials, and for the seasonal component is the trig functions or seasonal dummy variables.

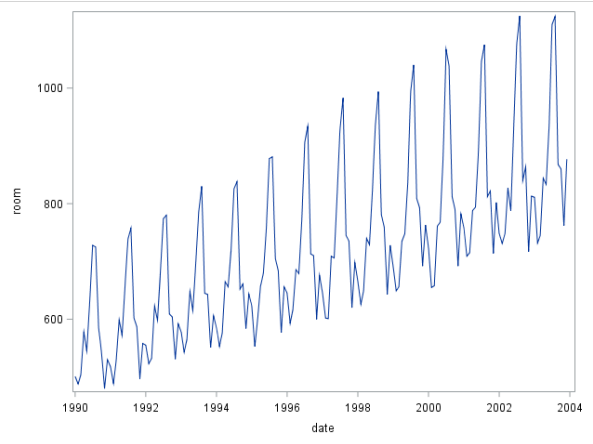
Recall the general linear model (GLM): Let $\{Y_t\}$ be a time series,

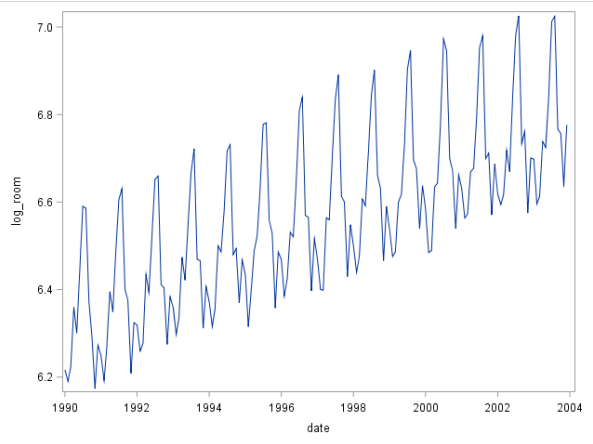
$$Y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_k x_{k,t} + \epsilon_t,$$

where ϵ_t are IID normal random variables.

In matrix notation:

Example 4.3. Consider the hotel example again. After taking the log, the series seem to have constant variance. Fit a regression model to estimate the trend and seasonal components.





5 Differencing

We have introduced estimation of the trend and seasonal components. An alternative approach is to remove the trend and seasonal components instead of estimating them. We would need to **difference** the series.
The **backshift operator** is defined by

$$Bx_t = x_{t-1}.$$

It can be extended to powers: $B^2x_t = B(Bx_t) = Bx_{t-1} = x_{t-2}$. Thus,

$$B^kx_t = x_{t-k}.$$

Furthermore, we may write

$$\nabla x_t = (1 - B)x_t = x_t - x_{t-1},$$

and define ∇ to be the **difference operator**.

Difference of order d is defined as

$$\nabla^d = (1 - B)^d$$

A linear trend can be removed by taking difference of order 1. Similarly, a polynomial trend of order k can be removed by taking difference of order k.

A seasonal component of period T can be removed by taking difference at lag T:

$$\nabla_T = (1 - B^T)$$

6 Estimation v.s. Differencing

7 General Approach to Univariate Time Series Analysis